Modul 08 Basis Data

PRAKTIKUM BASIS DATA

Scripting SQL (Create Index, Create View, Analyze Table)

DISUSUN OLEH:

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Jurnal 08

1. Buatlah desain database untuk suatu perusahaan:

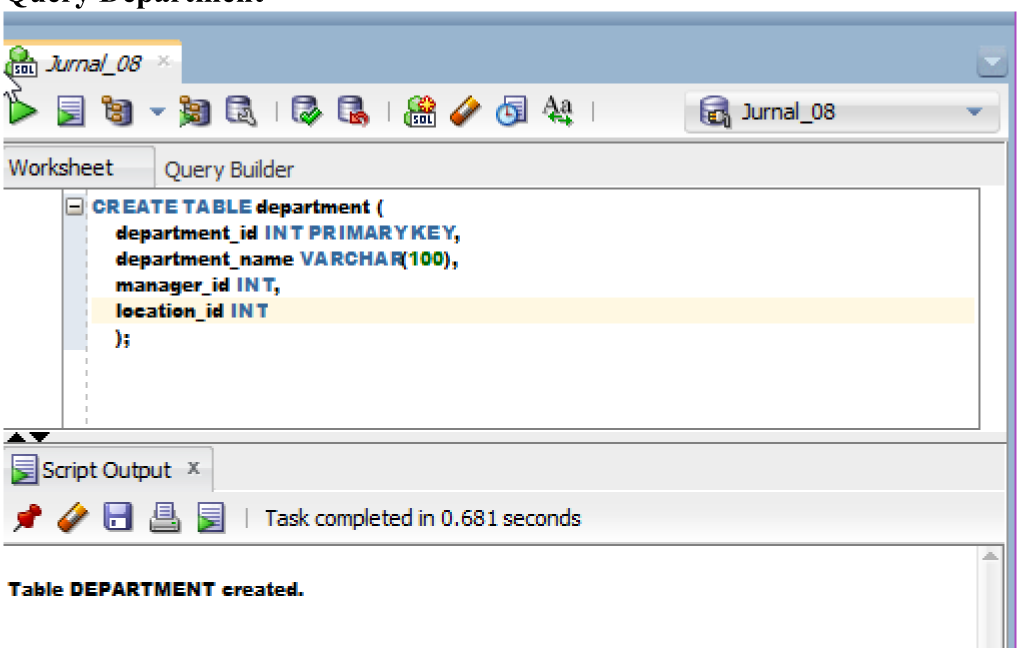
a. Table department dengan primary key adalah department_id. Tabel departemen terdiri atas field-field: department_id, department_name, manager_id, dan location_id.

b. Table employee dengan contraint foreign key adalah department_id. Tabel employee terdiri atas field-field: employee_id, last_nama NOT NULL, email, salary, commission_pct, hire_date NOT NULL>

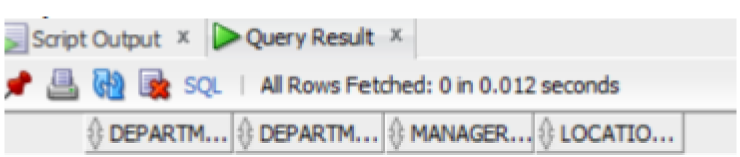
c. Tambahkan tabel-tabel lain yang saling berelasi (seperti: tabel persediaan barang, pemasok barang, dst).

Jawaban:

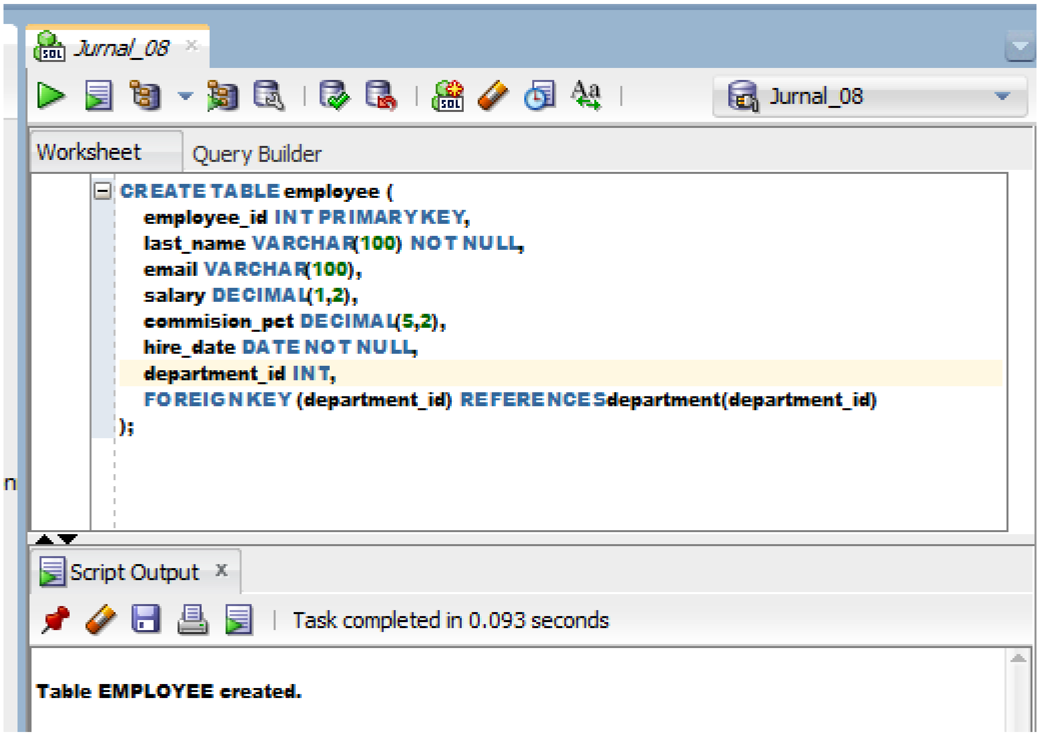
a. **Query Department**



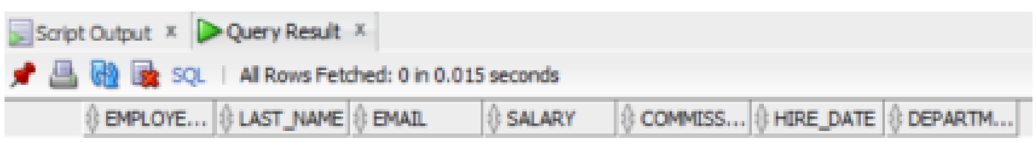
Output



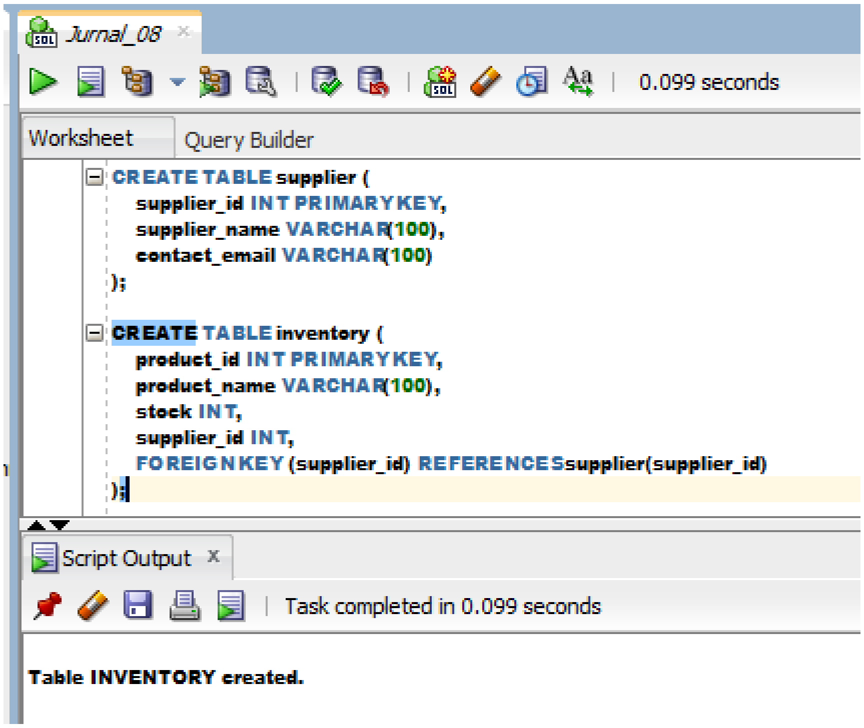
b. Query Employee



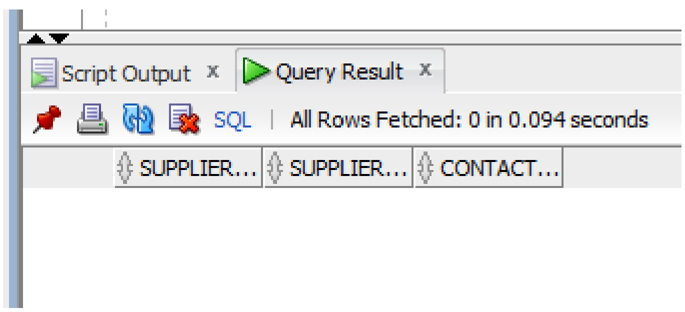
OUTPUT

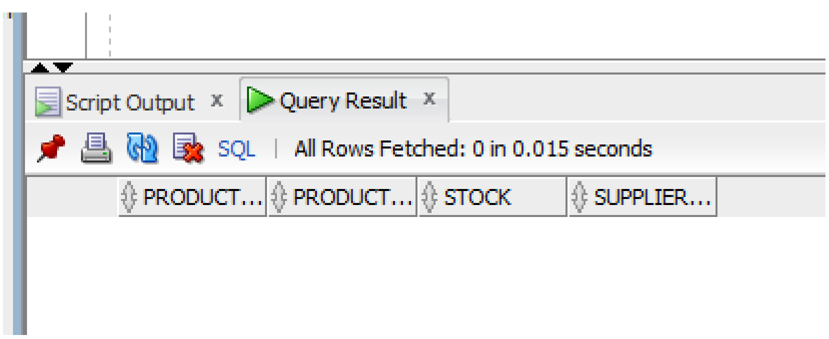


c. Query Supplier and Inventory



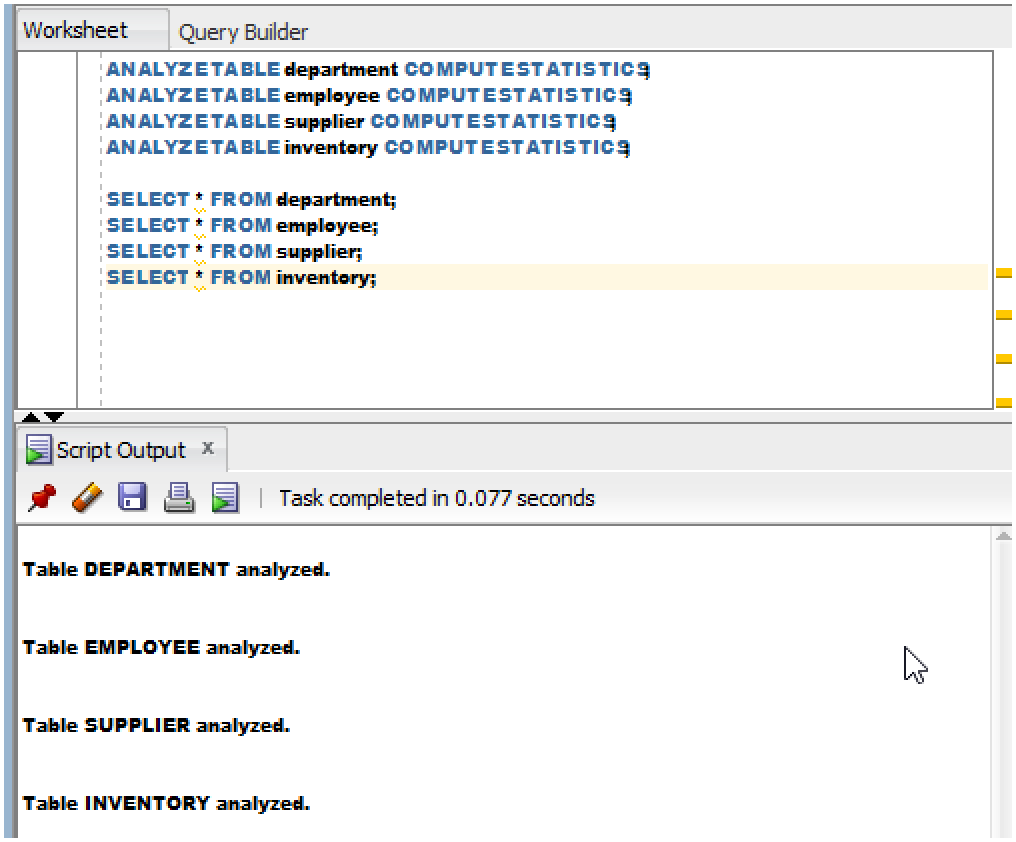
OUTPUT



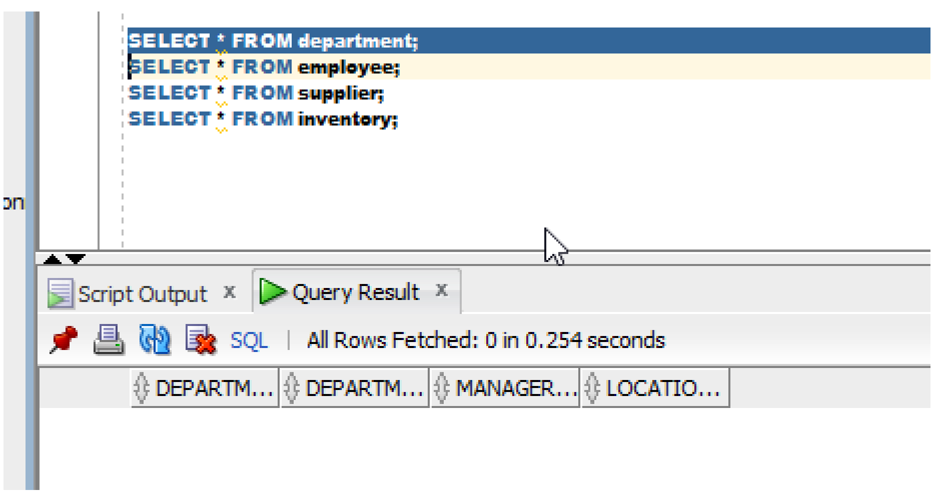


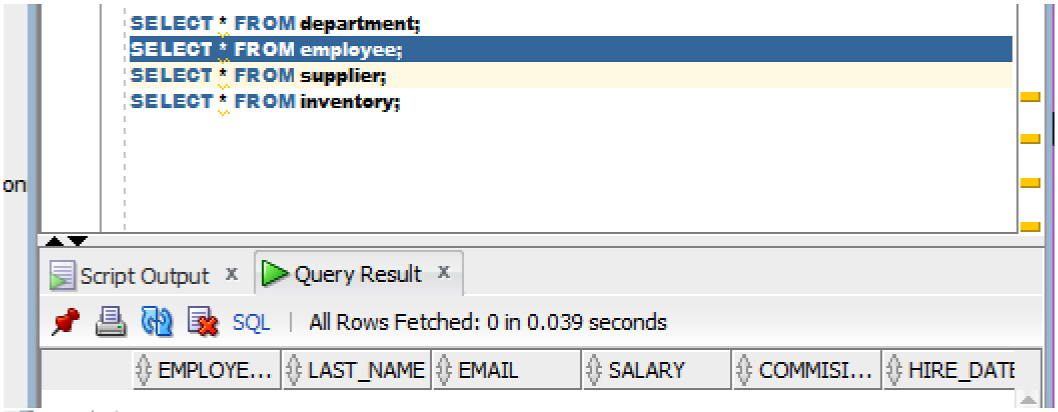
2. Analisis table-table dengan compute statistics

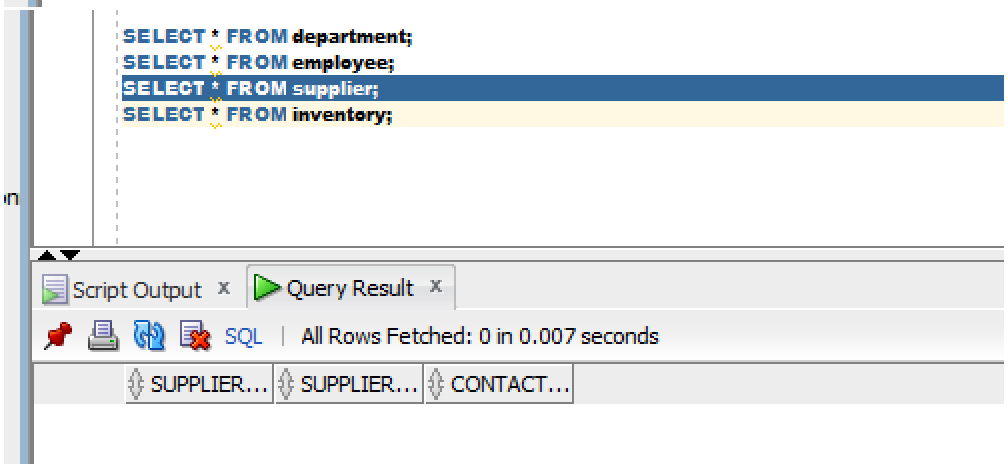
Query

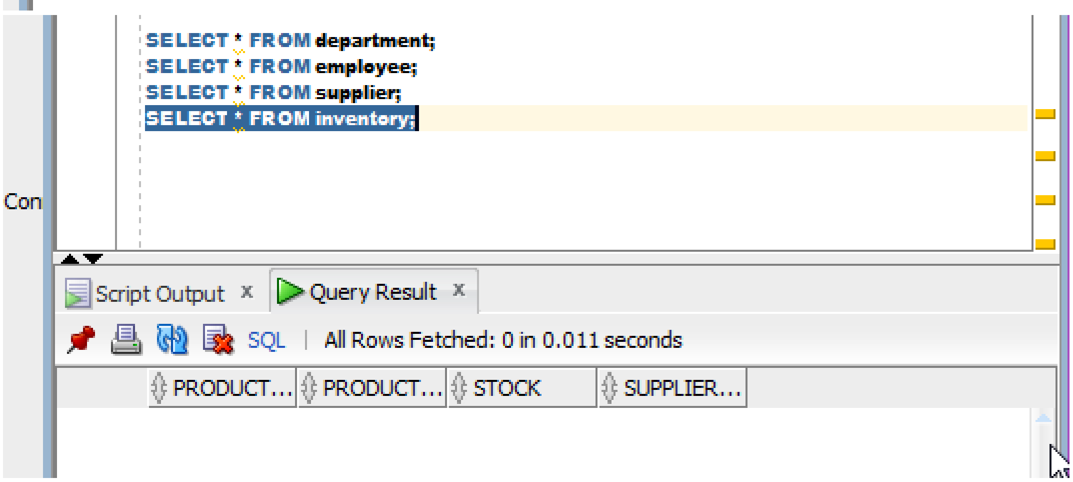

Worksheet
Query Builder
ANALYZE TABLE department COMPUTE STATISTICS;
ANALYZE TABLE employee COMPUTE STATISTICS;
ANALYZE TABLE supplier COMPUTE STATISTICS;
ANALYZE TABLE inventory COMPUTE STATISTICS;
SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM inventory;
Script Output
Task completed in 0.077 seconds
Table DEPARTMENT analyzed.
Table EMPLOYEE analyzed.
Table SUPPLIER analyzed.
Table INVENTORY analyzed.

OUTPUT


SELECT * FROM department;
SELECT * FROM employee;
SELECT * FROM supplier;
SELECT * FROM inventory;
Script Output
Query Result
SQL
All Rows Fetched: 0 in 0.254 seconds
DEPARTM...
DEPARTM...
MANAGER...
LOCATIO...







3. Isikan data data yang bersesuaian dengan tabel-tabel pada nomor 1 (masing masing 10 record)

Query Department

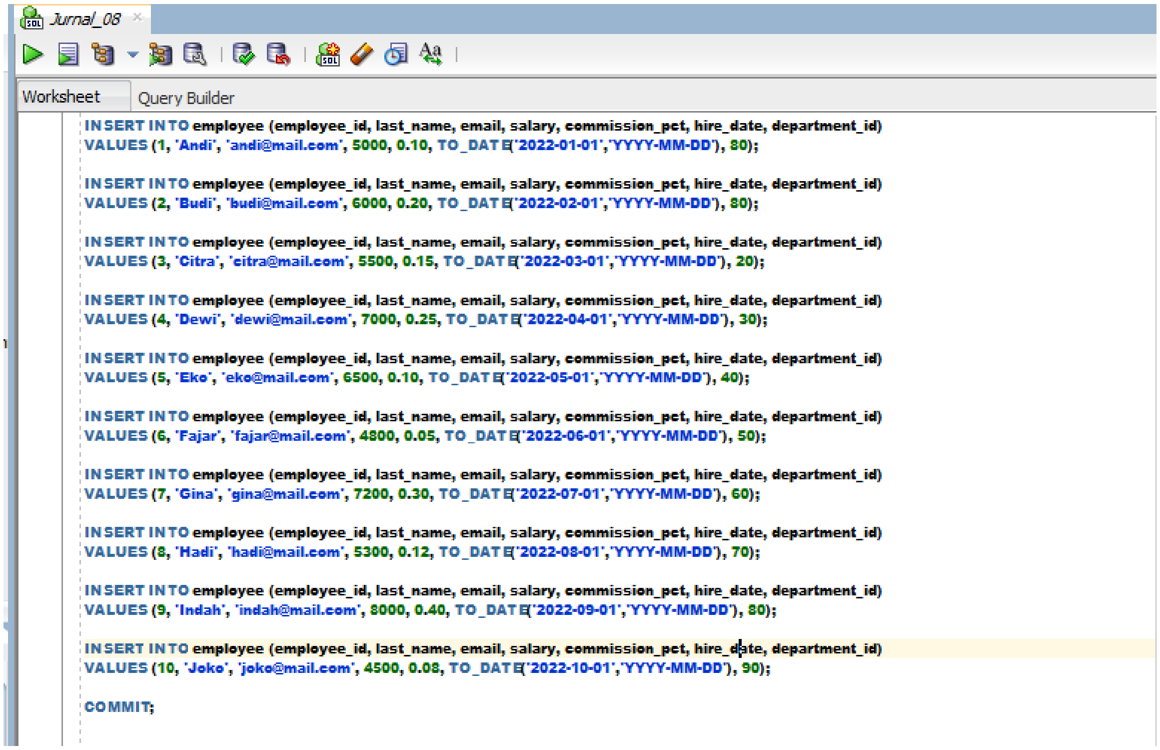
```sql
INSERT INTO department VALUES (10, 'HR', 101, 1001);
INSERT INTO department VALUES (20, 'IT', 102, 1002);
INSERT INTO department VALUES (30, 'Finance', 103, 1003);
INSERT INTO department VALUES (40, 'Marketing', 104, 1004);
INSERT INTO department VALUES (50, 'Sales', 105, 1005);
INSERT INTO department VALUES (60, 'Logistics', 106, 1006);
INSERT INTO department VALUES (70, 'Support', 107, 1007);
INSERT INTO department VALUES (80, 'Engineering', 108, 1008);
INSERT INTO department VALUES (90, 'QA', 109, 1009);
INSERT INTO department VALUES (100, 'Admin', 110, 1010);

COMMIT;
```

OUTPUT

All Rows Fetched: 10 in 0.062 seconds

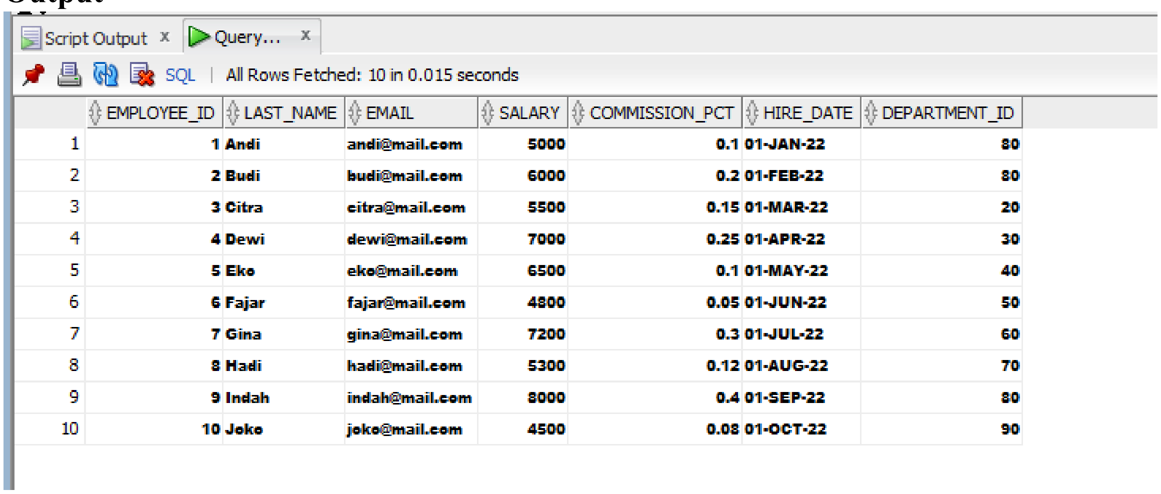
	DEPARTMENT_ID	DEPARTMENT_NAME	MANAGER_ID	LOCATION_ID
1	10	HR	101	1001
2	20	IT	102	1002
3	30	Finance	103	1003
4	40	Marketing	104	1004
5	50	Sales	105	1005
6	60	Logistics	106	1006
7	70	Support	107	1007
8	80	Engineering	108	1008
9	90	QA	109	1009
10	100	Admin	110	1010

Query Employee



```sql
INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (1, 'Andi', 'andi@mail.com', 5000, 0.10, TO_DATE('2022-01-01','YYYY-MM-DD'), 80);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (2, 'Budi', 'budi@mail.com', 6000, 0.20, TO_DATE('2022-02-01','YYYY-MM-DD'), 80);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (3, 'Citra', 'citra@mail.com', 5500, 0.15, TO_DATE('2022-03-01','YYYY-MM-DD'), 20);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (4, 'Dewi', 'dewi@mail.com', 7000, 0.25, TO_DATE('2022-04-01','YYYY-MM-DD'), 30);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (5, 'Eko', 'eko@mail.com', 6500, 0.10, TO_DATE('2022-05-01','YYYY-MM-DD'), 40);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (6, 'Fajar', 'fajar@mail.com', 4800, 0.05, TO_DATE('2022-06-01','YYYY-MM-DD'), 50);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (7, 'Gina', 'gina@mail.com', 7200, 0.30, TO_DATE('2022-07-01','YYYY-MM-DD'), 60);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (8, 'Hadi', 'hadi@mail.com', 5300, 0.12, TO_DATE('2022-08-01','YYYY-MM-DD'), 70);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (9, 'Indah', 'indah@mail.com', 8000, 0.40, TO_DATE('2022-09-01','YYYY-MM-DD'), 80);

INSERT INTO employee (employee_id, last_name, email, salary, commission_pct, hire_date, department_id)
VALUES (10, 'Joko', 'joko@mail.com', 4500, 0.08, TO_DATE('2022-10-01','YYYY-MM-DD'), 90);

COMMIT;
```

Output

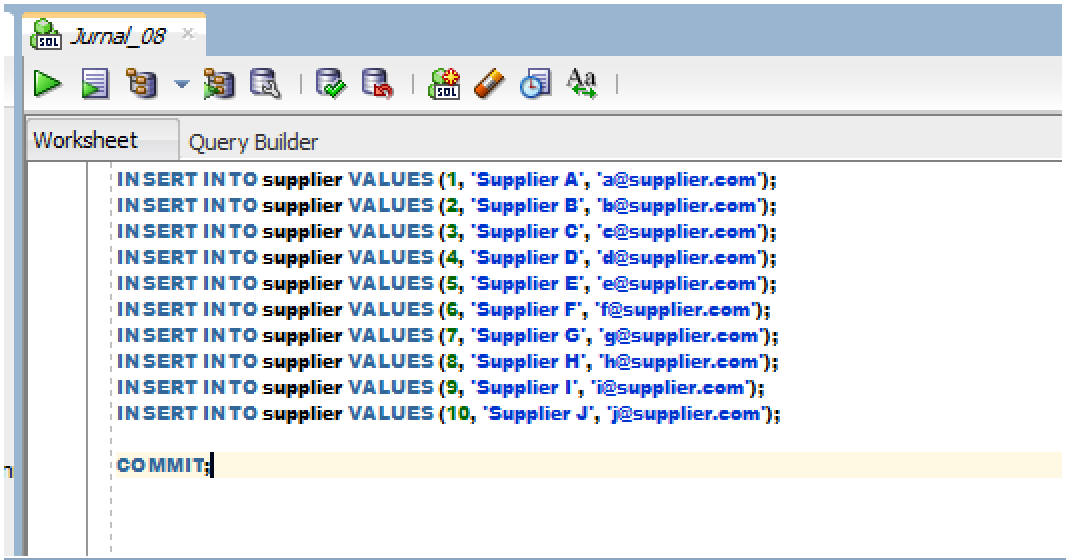


All Rows Fetched: 10 in 0.015 seconds

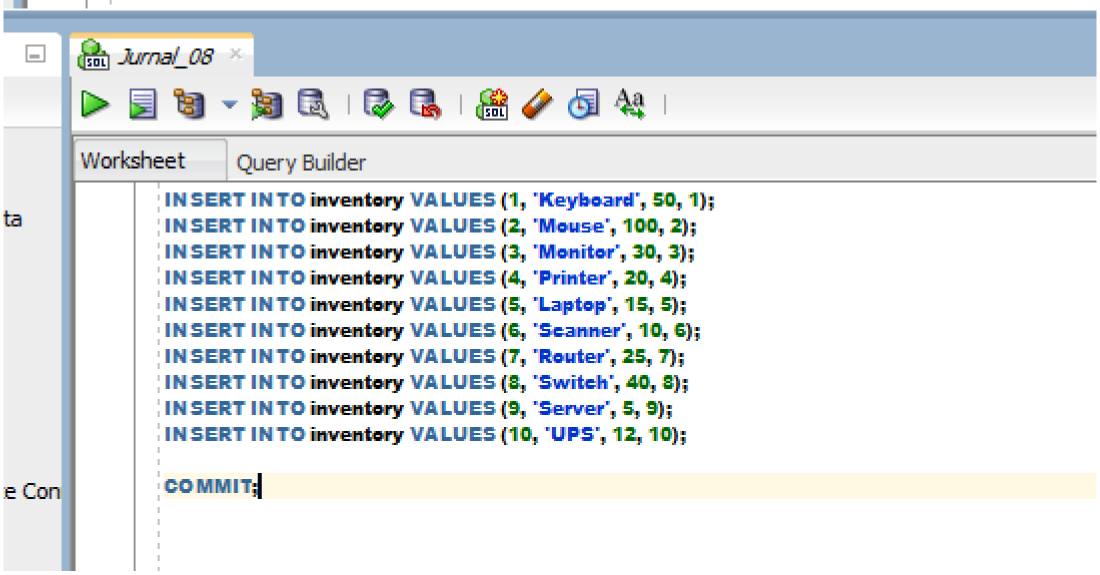
	EMPLOYEE_ID	LAST_NAME	EMAIL	SALARY	COMMISSION_PCT	HIRE_DATE	DEPARTMENT_ID
1	1	Andi	andi@mail.com	5000	0.1	01-JAN-22	80
2	2	Budi	budi@mail.com	6000	0.2	01-FEB-22	80
3	3	Citra	citra@mail.com	5500	0.15	01-MAR-22	20
4	4	Dewi	dewi@mail.com	7000	0.25	01-APR-22	30
5	5	Eko	eko@mail.com	6500	0.1	01-MAY-22	40
6	6	Fajar	fajar@mail.com	4800	0.05	01-JUN-22	50
7	7	Gina	gina@mail.com	7200	0.3	01-JUL-22	60
8	8	Hadi	hadi@mail.com	5300	0.12	01-AUG-22	70
9	9	Indah	indah@mail.com	8000	0.4	01-SEP-22	80
10	10	Joko	joko@mail.com	4500	0.08	01-OCT-22	90

Query Supplier & Inventory

QUERY




```sql
INSERT INTO supplier VALUES (1, 'Supplier A', 'a@supplier.com');
INSERT INTO supplier VALUES (2, 'Supplier B', 'b@supplier.com');
INSERT INTO supplier VALUES (3, 'Supplier C', 'c@supplier.com');
INSERT INTO supplier VALUES (4, 'Supplier D', 'd@supplier.com');
INSERT INTO supplier VALUES (5, 'Supplier E', 'e@supplier.com');
INSERT INTO supplier VALUES (6, 'Supplier F', 'f@supplier.com');
INSERT INTO supplier VALUES (7, 'Supplier G', 'g@supplier.com');
INSERT INTO supplier VALUES (8, 'Supplier H', 'h@supplier.com');
INSERT INTO supplier VALUES (9, 'Supplier I', 'i@supplier.com');
INSERT INTO supplier VALUES (10, 'Supplier J', 'j@supplier.com');

COMMIT;
```




```sql
INSERT INTO inventory VALUES (1, 'Keyboard', 50, 1);
INSERT INTO inventory VALUES (2, 'Mouse', 100, 2);
INSERT INTO inventory VALUES (3, 'Monitor', 30, 3);
INSERT INTO inventory VALUES (4, 'Printer', 20, 4);
INSERT INTO inventory VALUES (5, 'Laptop', 15, 5);
INSERT INTO inventory VALUES (6, 'Scanner', 10, 6);
INSERT INTO inventory VALUES (7, 'Router', 25, 7);
INSERT INTO inventory VALUES (8, 'Switch', 40, 8);
INSERT INTO inventory VALUES (9, 'Server', 5, 9);
INSERT INTO inventory VALUES (10, 'UPS', 12, 10);

COMMIT;
```

OUTPUT

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Worksheet | Query Builder

```sql
SELECT * FROM supplier;
SELECT * FROM inventory;
```

Script Output | Query Result

SQL | All Rows Fetched: 10 in 0.014 seconds

	SUPPLIER_ID	SUPPLIER_NAME	CONTACT_EMAIL
1	1	Supplier A	a@supplier.com
2	2	Supplier B	b@supplier.com
3	3	Supplier C	c@supplier.com
4	4	Supplier D	d@supplier.com
5	5	Supplier E	e@supplier.com
6	6	Supplier F	f@supplier.com
7	7	Supplier G	g@supplier.com
8	8	Supplier H	h@supplier.com
9	9	Supplier I	i@supplier.com
10	10	Supplier J	j@supplier.com

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Worksheet | Query Builder

```sql
SELECT * FROM supplier;
SELECT * FROM inventory;
```

Script Output | Query Result

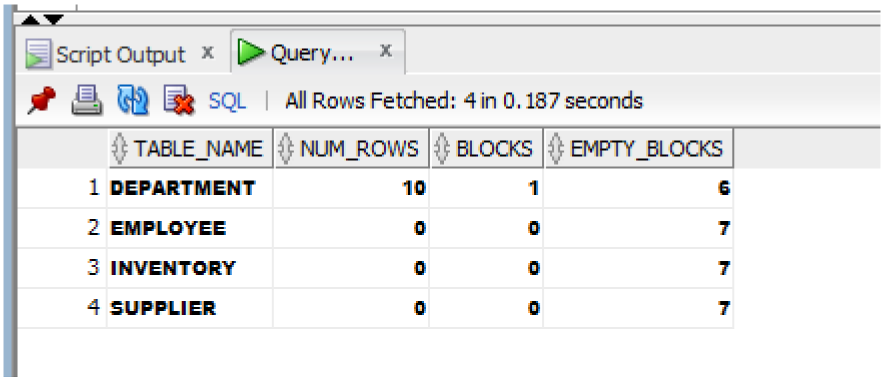
SQL | All Rows Fetched: 10 in 0.008 seconds

	PRODUCT_ID	PRODUCT_NAME	STOCK	SUPPLIER_ID
1	1	Keyboard	50	1
2	2	Mouse	100	2
3	3	Monitor	30	3
4	4	Printer	20	4
5	5	Laptop	15	5
6	6	Scanner	10	6
7	7	Router	25	7
8	8	Switch	40	8
9	9	Server	5	9
10	10	UPS	12	10

4. Analisis table-table dengan compute statistics

Jawaban

Output Departement, Employee, supplier and inventory



	TABLE_NAME	NUM_ROWS	BLOCKS	EMPTY_BLOCKS
1	DEPARTMENT	10	1	6
2	EMPLOYEE	0	0	7
3	INVENTORY	0	0	7
4	SUPPLIER	0	0	7

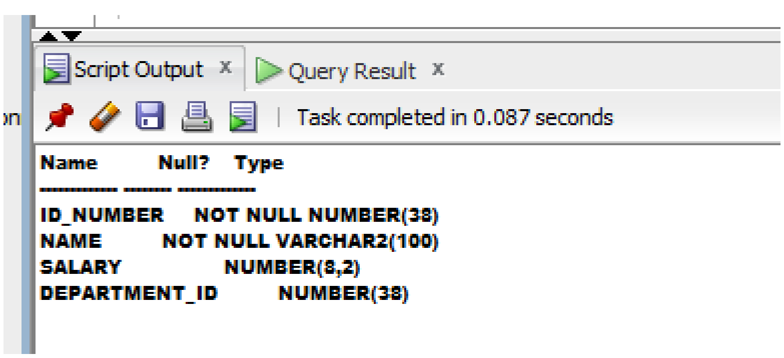
5. Buat view empvu80 yang berisi id_number, name, salary, department_id dari pegawai yang bekerja di department = 80.

Jawaban

```
CREATE VIEW empvu80 AS
SELECT
    employee_id AS id_number,
    last_name AS name,
    salary,
    department_id
FROM employee
WHERE department_id = 80;
```

6. Kemudian tampilkan struktur dari view empvu80.

Jawaban

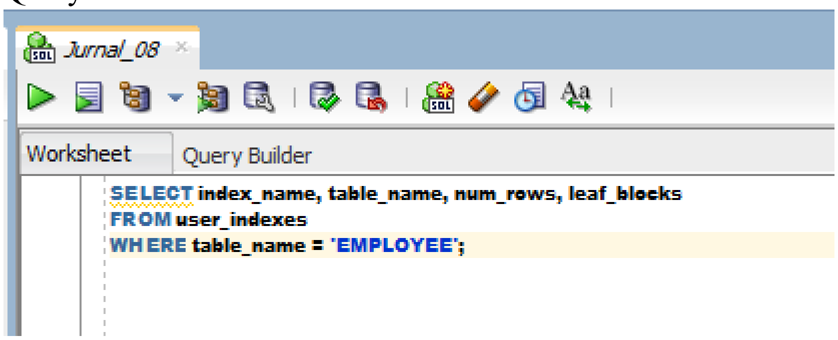



```
Name          Null?    Type
------------- -------- ------------
ID_NUMBER     NOT NULL NUMBER(38)
NAME          NOT NULL VARCHAR2(100)
SALARY                 NUMBER(8,2)
DEPARTMENT_ID          NUMBER(38)
```

7. Buat non-unique index pada kolom FOREIGN KEY yang ada pada tabel EMPLOYEE.

Jawaban

Query



Output